

RESEARCH ARTICLE

Prognostic value of plasma %p-tau217 in cognitively unimpaired older adults

Jesús Silva-Rodríguez^{1,2} | Linda Zhang¹ | Luca Kleineidam^{3,4} |
 Francisco J. López-González¹ | Sonia Wagner-Reguero¹ | Minerva Martínez-Castillo¹ |
 Aitana Sogorb-Esteve^{1,5} | Michelle E. Guamán-Chulunchana¹ | Ana B. Pastor¹ |
 Nekane Moreno¹ | Andrea Ruiz-Calvo¹ | Cristina Sánchez-Martín¹ | Eva Alfayate¹ |
 Marta Molero-Cartón¹ | Elizabeth L. Valeriano-Lorenzo¹ | Belén Frades-Payo¹ |
 Meritxell Valentí¹ | María A. Zea-Sevilla¹ | Alexis Moscoso^{6,7} |
 Alfredo Ramírez^{3,4,8,9,10} | Pamela Martino-Adami⁸ | Steffi G. Riedel-Heller¹¹ |
 Martin Scherer¹² | Andreas Jeromin¹³ | Miguel Calero^{1,2,14} | Teodoro del Ser¹ |
 Anja Schneider^{3,4} | Michel J. Grothe^{1,2} | Pascual Sánchez-Juan^{1,2}

¹Reina Sofía Foundation Alzheimer Centre, CIEN Foundation, ISCIII, Madrid, Spain²Centro de Investigación Biomédica en Red sobre Enfermedades Neurodegenerativas (CIBERNED), Madrid, Spain³Department of Old Age Psychiatry and Cognitive Disorders, University Hospital Bonn, Bonn, Germany⁴German Center for Neurodegenerative Diseases (DZNE), Bonn, Germany⁵Queen Square Institute of Neurology, University College London, London, UK⁶Nuclear Medicine Department and Molecular Imaging Group, Instituto de Investigación Sanitaria de Santiago de Compostela, Santiago de Compostela, Spain⁷Wallenberg Center for Molecular and Translational Medicine and Department of Psychiatry and Neurochemistry, University of Gothenburg, Gothenburg, Sweden⁸Division of Neurogenetics and Molecular Psychiatry Department of Psychiatry and Psychotherapy, Faculty of Medicine and University Hospital Cologne, University of Cologne, Cologne, Germany⁹Department of Psychiatry, Glenn Biggs Institute for Alzheimer's and Neurodegenerative Diseases, San Antonio, Texas, USA¹⁰Cluster of Excellence Cellular Stress Responses in Aging-Associated Diseases (CECAD), University of Cologne, Cologne, Germany¹¹Institute of Social Medicine, Occupational Health and Public Health, University of Leipzig, Leipzig, Germany¹²Department of Primary Medical Care, Center for Psychosocial Medicine, University Medical Center, Hamburg-Eppendorf, Hamburg, Germany¹³Tau Biosciences, Inc. Carlsbad, California, USA¹⁴Chronic Disease Programme (UFIEC), Instituto de Salud Carlos III, Madrid, Spain

Correspondence

Pascual Sánchez-Juan and Michel Grothe,
 Fundación CIEN, Centro Alzheimer Reina
 Sofía; C/ Valderrebollo, 5, 28031, Madrid,
 Spain.

Abstract

INTRODUCTION: We evaluated the prognostic value of baseline plasma phosphorylated tau 217 ratio (%p-tau217) for predicting long-term progression in cognitively unimpaired (CU) older adults.

Jesús Silva-Rodríguez, Linda Zhang, and Luca Kleineidam contributed equally to the study and share first authorship

Michel J. Grothe and Pascual Sánchez-Juan contributed equally to the study and share senior authorship

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2026 The Author(s). *Alzheimer's & Dementia* published by Wiley Periodicals LLC on behalf of Alzheimer's Association.

Finally, the association between p-tau217 and AD-related neurodegeneration is still insufficiently characterized.¹⁹ Clarifying this link is essential, as neurodegeneration is a core pathophysiological hallmark of the disease and closely correlates with clinical progression.¹ Here, we leveraged data from a large, single-center cohort of older CU individuals, with up to 12 years of annual follow-ups, to assess the prognostic value of baseline plasma %p-tau217 for long-term clinical progression, cognitive decline, and neurodegeneration in this population. Findings were replicated using immunoassay-based p-tau217 in the same and in a primary care cohort, another important target population for future screening.

2 | METHODS

2.1 | Study participants

The Vallecas Project (VP) is a longitudinal, single-center, community-based study conducted in Madrid, Spain. This study enrolled 1173 older individuals (age range: 69 to 85 years; 64% female) who self-perceived as physically and cognitively healthy. Participants underwent annual clinical and neuropsychological assessment, blood extraction, and 3T magnetic resonance imaging (MRI), spanning a total of 12 years (2011 to 2023). Detailed inclusion and exclusion criteria are provided in Supplementary Methods 1.1. All participants provided written informed consent, and the VP was approved by the ethics committee of the Instituto de Salud Carlos III. For this study, we included individuals with sufficient plasma available for LC-MS/MS analysis at or within 2 years of their initial visit, which served as baseline for subsequent analyses, and whose CU status at that visit was confirmed using standardized criteria.

2.2 | Clinical and neuropsychological assessment

At each annual visit participants completed a semi-structured medical interview and a comprehensive neuropsychological assessment evaluating multiple cognitive domains, subjective cognitive decline (SCD), and functional impairment (Supplementary Methods 1.2.1). After each visit, a clinical diagnosis of CU, mild cognitive impairment (MCI), or dementia was assigned by consensus between the clinical and neuropsychological evaluators, based on standardized criteria (Supplementary Methods 1.2.2).

2.3 | Plasma biomarker measurements

Blood samples were collected annually and processed as detailed in Supplementary Materials 1.3.1. The plasma analyses were performed while all personnel analyzing the samples were blinded to all clinical or biomarker data. LC-MS/MS, performed by C2N Diagnostics, LLC (St. Louis, MO, USA), was used to quantify baseline plasma levels of

RESEARCH IN CONTEXT

- Systematic review:** We searched PubMed from inception through July 7, 2025, for studies investigating longitudinal outcomes associated with p-tau217 in cognitively unimpaired individuals. Several studies reported that elevated plasma p-tau217 levels were associated with tau and amyloid accumulation as assessed by PET and CSF. However, evidence remains limited regarding the capacity of plasma p-tau217 biomarkers to identify cognitively unimpaired individuals at substantial long-term risk of progression to MCI or dementia.
- Interpretation:** Elevated plasma %p-tau217 robustly associated with accelerated cognitive decline, increased risk of progression to MCI and dementia, and faster neurodegeneration. High negative predictive values (>90%) suggest that p-tau217 may be an excellent tool for identifying low-risk community individuals. Overall, our results support incorporating plasma p-tau217 testing into risk-stratification strategies for preclinical AD in both clinical practice and research.
- Future directions:** Future studies may validate our findings in more diverse populations, including younger individuals, different demographics, and diverse genetic backgrounds.

p-tau217 (p-tau217_{C2N}), np-tau217 (np-tau217_{C2N}), and %p-tau217 (Supplementary Methods 1.3.2).²⁰

Longitudinal plasma levels of glial fibrillary acidic protein (GFAP) and neurofilament light chain (NfL) were quantified locally using the SIMOA platform (Quanterix, Billerica, MA, USA)²¹ at up to three time points, selected to maximize follow-up coverage, for a subset of randomly selected subjects ($n = 487$ Supplementary Methods 1.3.3).

2.4 | MRI imaging

Three-dimensional T1-weighted MRI scans were acquired at each visit using a 3T MRI scanner (GE Signa HDxt, Wisconsin, USA). Serial measurements of hippocampal volume (HippVol) were derived using longitudinal volumetry as implemented within the Computational Anatomy Toolbox version 12.9.²² Additional details are provided in Supplementary Materials 1.4.

2.5 | Outcomes

The primary outcomes of our study were as follows: (1) cognitive trajectories of participants with different levels of baseline %p-tau217, assessed using a modified version of the Preclinical Alzheimer's Cog-

nitive Composite Scale (mPACC, Supplementary Methods 1.2.3)²³ and the Mini-Mental State Examination (MMSE); (2) clinical outcomes associated with different levels of baseline %p-tau217, including (a) longitudinal trajectories of clinical and functional deterioration, assessed using the Clinical Dementia Rating Scale sum of boxes (CDR-SB) and the Functional Activities Questionnaire (FAQ), and (b) individual risks of clinical progression to MCI or dementia; and (3) trajectories of neurodegeneration and neuroinflammation markers, consisting of (a) medial temporal lobe atrophy as indicated by HippVol; (b) neuronal injury, as indicated by plasma NFL; and (c) astrocyte reactivity and damage, as measured by plasma GFAP.

2.6 | Statistical analysis

Statistical analyses were conducted using R version 4.4.2. Longitudinal trajectories of the outcome variables were modeled using generalized additive models (GAMs), with baseline %p-tau217 as the main predictor and with age at baseline, sex, years of education, and APOE ϵ 4 carriership included as covariates. %p-tau217 was treated as a categorical variable based on a three cut-off classification: Low (<2.66%), Intermediate (2.66 to 4.90%), Elevated (4.90 to 7.40%), and High ($\geq$ 7.40%), where 2.66% and 4.90% corresponded to previously derived optimal cut-offs for predicting amyloid and tau PET positivity, respectively,²⁴ while $\geq$ 7.40% served as a high-risk cut-off intended to capture individuals with higher likelihood of advanced tau pathology.²⁵ Importantly, the use of predefined, biologically informed thresholds rather than data-driven quantiles provides a more realistic approach for assessing the potential performance of %p-tau217 in real-world populations, where fixed decision boundaries are essential for interpretation and implementation. GAMs included a general non-linear smooth term to model the potential temporal non-linearity of the variable trajectory and a “by %p-tau217” interaction term accounting for differential temporal trajectories across the different %p-tau217 groups. Finally, subject-specific random intercepts and slopes were included. We report comparisons between groups at baseline, 5 years, and 10 years of follow-up. To quantify the proportion of variance in longitudinal trajectories uniquely attributable to plasma %p-tau217, we compared full and null population-average GAMs that differed only in the inclusion of %p-tau217 group, both as a main effect and as a group-specific time smooth. The variance contribution of the biomarker was estimated as the difference in deviance explained between models, expressed as a percentage. Additional details about the GAM analysis are provided in Supplementary Materials 3.1.

Hazard ratios (HRs) for the progression to MCI and/or dementia were estimated using interval-censored Cox regression models, including the same covariates as the GAM models. The positive predictive value (PPV) and negative predictive value (NPV) for predicting MCI/dementia progression of the different %p-tau217 cut-offs were estimated based on inverse probability of censoring weights (IPCW) at 5 and 10 years after baseline (Supplementary Materials 3.2). HRs, NPVs, and PPVs are reported together with 95% confidence intervals (CIs).

2.7 | Replication analyses

Complementary p-tau217 measurements for the VP were obtained locally using a chemiluminescent enzyme immunoassay available for the automated LUMIPULSE platform from Fujirebio (p-tau217_{Fujirebio}, Supplementary Methods 1.3.4) and used to replicate the main outcomes described earlier.

Additionally, we report data from 1204 CU participants enrolled in the German Study on Ageing, Cognition and Dementia (AgeCoDe), a prospective multicenter study involving primary care patients aged over 80 years and following a study protocol and follow-up schedule comparable to that of the VP. Plasma p-tau217 measurements were obtained using the SIMOA ALZPath immunoassay platform (p-tau217_{ALZPath}) and were used to replicate (1) trajectories of global cognition (based on a comparable global cognitive composite, mPACC*, and MMSE); (2) trajectories of clinical (CDR-SB) and functional decline, as measured by the Lawton and Brody Instrumental Activities of Daily Living scale (IADL), used as a proxy for FAQ; and (3) individual risks of clinical progression to MCI/dementia at 5 and 7.5 years. Methodological details of the AgeCoDe are provided in Supplementary Methods 2.

For replication analyses, participants were first stratified according to baseline p-tau217 using previously derived optimal cut-offs for detecting amyloid and tau PET positivity, respectively, for both p-tau217_{Fujirebio} and p-tau217_{ALZPath}.^{9,26} To match the three-cut-off strategy used for %p-tau217, an additional cut-off was applied to define a High group comprising the same percentile of individuals with the highest p-tau217_{Fujirebio} (VP) or p-tau217_{ALZPath} (AgeCoDe) values as the %p-tau217 High group.

3 | RESULTS

The study included 982 CU individuals, following exclusion of 49 participants who were classified as having MCI at baseline, 130 with insufficient plasma available for LC-MS/MS, and 12 whose LC-MS/MS results did not meet quality control criteria. Participants were categorized as Low ($n = 674$, 68.6%), Intermediate ($n = 145$, 14.8%), Elevated ($n = 103$, 10.5%), and High ($n = 60$, 6.1%). Baseline characteristics of the cohort stratified by %p-tau217 group are presented in Table 1. Figure S1 provides pairwise comparisons of the most relevant outcomes.

3.1 | Longitudinal trajectories of cognition across %p-tau217 groups

mPACC and MMSE trajectories significantly differed across all %p-tau217 groups (Figure 1A,B), with higher %p-tau217 levels linked to progressively steeper declines in both metrics ($p < 0.001$ for all smooth difference terms). The Low and Intermediate groups demonstrated minimal decline over the study duration ($\Delta_{\text{mPACC}} < 0.58$ points), even displaying a mild initial practice effect indicated by increasing mPACC scores in early visits. The Elevated group did not differ from

Needs, Health Service Use, Costs, and Health-Related Quality of Life in a Large Sample of Oldest-Old Primary Care Patients [85+; AgeQualiDe]) and was funded by the German Federal Ministry of Education and Research (grants KND: 01GI0102, 01GI0420, 01GI0422, 01GI0423, 01GI0429, 01GI0431, 01GI0433, 01GI0434; grants KNDD: 01GI0710, 01GI0711, 01GI0712, 01GI0713, 01GI0714, 01GI0715, 01GI0716; grants Health Service Research Initiative: 01GY1322A, 01GY1322B, 01GY1322C, 01GY1322D, 01GY1322E, 01GY1322F, and 01GY1322G). Further funding for biomarker measurement was obtained from the German Federal Ministry of Education and Research (DESCARTES consortium, FZK 01EK2102A; PREPARE, FZK 01GP2213A) and the Alzheimer Forschung Initiative (Grant number: #20065p).

CONFLICT OF INTEREST STATEMENT

Jesús Silva-Rodríguez is a consultant, stakeholder, and board member of QubioTech Health Intelligence. A.M. is an employee of ZRO Imaging AB. Andreas Jeromin is a former employee of ALZ path, Inc., holds stock options, and is a co-founder of Tau Biosciences, Inc., in which he holds stock options. All other authors declare no competing interests. Author disclosures are available in the [Supporting Information](#).

CONSENT STATEMENT

Written informed consent was obtained for all participants prior to being included in the study. The study was approved by the corresponding ethics committees of the respective institutions. The study was conducted in accordance with the Declaration of Helsinki and applicable local regulations.

REFERENCES

- Jack CR, Andrews JS, Beach TG, et al. Revised criteria for diagnosis and staging of Alzheimer's disease: Alzheimer's association workgroup. *Alzheimers Dement*. Published online 2024;20(8):5143-5169. <https://doi.org/10.1002/alz.13859>
- Villain N, Planché V, Lilamand M, et al. Lecanemab for early Alzheimer's disease: Appropriate use recommendations from the French federation of memory clinics. *J Prev Alzheimers Dis*. 2025;12(4):100094. <https://doi.org/10.1016/j.tjpad.2025.100094>
- Yaari R, Holdridge KC, Williamson M, et al. Donanemab in preclinical Alzheimer's disease: Screening and baseline data from TRAILBLAZER-ALZ 3. *Alzheimers Dement*. 2025;21(9):e70662. <https://doi.org/10.1002/alz.70662>
- Sperling RA, Donohue MC, Raman R, et al. Trial of Solanezumab in preclinical Alzheimer's disease. *N Engl J Med*. 2023;389(12):1096-1107. <https://doi.org/10.1056/NEJMoa2305032>
- Aisen P, Bateman RJ, Crowther D, et al. The case for regulatory approval of amyloid-lowering immunotherapies in Alzheimer's disease based on clearcut biomarker evidence. *Alzheimers Dement*. 2025;21(1):e14342. <https://doi.org/10.1002/alz.14342>
- Therriault J, Vermeiren M, Servaes S, et al. Association of Phosphorylated Tau biomarkers with Amyloid Positron Emission Tomography vs Tau Positron Emission Tomography. *JAMA Neurol*. 2023;80(2):188. <https://doi.org/10.1001/jamaneurol.2022.4485>
- Karikari TK, Pascoal TA, Ashton NJ, et al. Blood phosphorylated tau 181 as a biomarker for Alzheimer's disease: a diagnostic performance and prediction modelling study using data from four prospective cohorts. *The Lancet Neurology*. 2020;19(5):422-433. [https://doi.org/10.1016/S1474-4422\(20\)30071-5](https://doi.org/10.1016/S1474-4422(20)30071-5)
- Montoliu-Gaya L, Alosco ML, Yhang E, et al. Optimal blood tau species for the detection of Alzheimer's disease neuropathology: An immunoprecipitation mass spectrometry and autopsy study. *Acta Neuropathol*. 2023;147(1):5. <https://doi.org/10.1007/s00401-023-02660-3>
- Ashton NJ, Brum WS, Di Molfetta G, et al. Diagnostic accuracy of a plasma Phosphorylated Tau 217 immunoassay for Alzheimer disease pathology. *JAMA Neurol*. Published online 2024;81(3):255-263. <https://doi.org/10.1001/jamaneurol.2023.5319>
- Ossenkoppele R, Salvadó G, Janelidze S, et al. Plasma p-tau217 and tau-PET predict future cognitive decline among cognitively unimpaired individuals: implications for clinical trials. *Nat Aging*. 2025;5(5):883-896. <https://doi.org/10.1038/s43587-025-00835-z>
- Fernández Arias J, Brum WS, Salvadó G, et al. Plasma phosphorylated tau217 strongly associates with memory deficits in the Alzheimer's disease spectrum. *Brain*. Published online 2025;148(7):2384-2399. <https://doi.org/10.1093/brain/awaf033>
- Meyer MR, Kirmess KM, Eastwood S, et al. Clinical validation of the PrecivityAD2 blood test: A mass spectrometry-based test with algorithm combining %p-tau217 and Aβ42/40 ratio to identify presence of brain amyloid. *Alzheimers Dement*. 2024;20(5):3179-3192. <https://doi.org/10.1002/alz.13764>
- Janelidze S, Barthélemy NR, He Y, Bateman RJ, Hansson O. Mitigating the Associations of Kidney Dysfunction With Blood Biomarkers of Alzheimer Disease by Using Phosphorylated Tau to Total Tau Ratios. *JAMA Neurol*. 2023;80(5):516. <https://doi.org/10.1001/jamaneurol.2023.0199>
- Warmenhoven N, Salvadó G, Janelidze S, et al. A comprehensive head-to-head comparison of key plasma phosphorylated tau 217 biomarker tests. *Brain*. 2025;148(2):416-431. <https://doi.org/10.1093/brain/awae346>
- Barthélemy NR, Salvadó G, Schindler SE, et al. Highly accurate blood test for Alzheimer's disease is similar or superior to clinical cerebrospinal fluid tests. *Nat Med*. 2024;30(4):1085-1095. <https://doi.org/10.1038/s41591-024-02869-z>
- Sperling RA, Donohue MC, Rissman RA, et al. Amyloid and Tau prediction of cognitive and functional decline in unimpaired older individuals: Longitudinal data from the A4 and LEARN Studies. *J Prev Alz Dis*. Published online 2024;11(4):802-813. doi:10.14283/jpad.2024.122
- Yakoub Y, Gonzalez-Ortiz F, Ashton NJ, et al. Plasma p-tau217 identifies cognitively normal older adults who will develop cognitive impairment in a 10-year window. *Alzheimers Dement*. 2025;21(2):e14537. <https://doi.org/10.1002/alz.14537>
- Grande G, Valletta M, Rizzuto D, et al. Blood-based biomarkers of Alzheimer's disease and incident dementia in the community. *Nat Med*. Published online 2025;31(6):2027-2035. <https://doi.org/10.1038/s41591-025-03605-x>
- Sanchez E, Coughlan GT, Wilkinson T, et al. Association of plasma biomarkers with longitudinal atrophy and microvascular burden on MRI across neurodegenerative and cerebrovascular diseases. *Neurology*. 2025;104(7):e213438. <https://doi.org/10.1212/WNL.0000000000213438>
- Eastwood SM, Meyer MR, Kirmess KM, et al. PrecivityAD2TM blood test: Analytical validation of an LC-MS/MS assay for quantifying plasma phospho-tau217 and non-phospho-tau217 peptide concentrations that are used with plasma amyloid-β42/40 in a multianalyte assay with algorithmic analysis for detecting brain amyloid pathology. *Diagnostics (Basel)*. 2024;14(16):1739. <https://doi.org/10.3390/diagnostics14161739>
- Chatterjee P, Pedrini S, Doecke JD, et al. Plasma Aβ42/40 ratio, p-tau181, GFAP, and NfL across the Alzheimer's disease continuum: A cross-sectional and longitudinal study in the AIBL cohort. *Alzheimers Dement*. 2023;19(4):1117-1134. <https://doi.org/10.1002/alz.12724>
- Gaser C, Dahnke R, Thompson PM, Kurth F, Luders E. The Alzheimer's Disease Neuroimaging Initiative null. CAT: a computational anatomy